FLATNESS AND REGULARITY OF THE QUOTIENT BY A SUBFIELD-GENERATED POLYNOMIAL SUBALGEBRA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Editorial note. Benign-screen note: the proof contains the phrase 'it is clear that' / 'is obvious' once, used as a legitimate step inside a complete argument; the author proves the statement, so this is not a hint-only proof.

Allowed standard prerequisites (editorial).

- Characterisation of geometric regularity through the naive cotangent complex. Proposition \ref{proposition-characterization-geometrically-regular}, cited at proof lines 8871-8872. Gives the isomorphism used for the subalgebra.
- Injectivity of the map on lower homology along the extension of residue fields. Lemma \ref{lemma-transitivity-gamma}, cited at proof lines 8884-8885. Supplies the injectivity in the commutative square.

Source reference. Original lemma 07E6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9436–9436. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07E5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9435–9435. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07E2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9432–9432. Navigation pointer to a statement the

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Source reference. Original lemma 07DY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9428–9428. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00NQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 868–868. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let k be a field of characteristic $p > 0$. Let $(A, \mathfrak{m}, K)$ be a Noetherian local k-algebra. Assume A is geometrically regular over k. Let $K/F/k$ be a finitely generated subextension. Let $\varphi : k[y_1, \ldots, y_m] \to A$ be a k-algebra map such that y_i maps to an element of F in K and such that $dy_1, \ldots, dy_m$ map to a basis of $\Omega_{F/k}$. Set $\mathfrak{p} = \varphi^{-1}(\mathfrak{m})$. Then*

$$k[y_1, \ldots, y_m]_{\mathfrak{p}} \to A$$

is flat and $A/\mathfrak{p}A$ is regular.

Proof. Set $A_0 = k[y_1, \ldots, y_m]_{\mathfrak{p}}$ with maximal ideal $\mathfrak{m}_0$ and residue field K_0. Note that $\Omega_{A_0/k}$ is free of rank m and $\Omega_{A_0/k} \otimes K_0 \to \Omega_{K_0/k}$ is an isomorphism. It is clear that A_0 is geometrically regular over k. Hence $H_1(L_{K_0/k}) \to \mathfrak{m}_0/\mathfrak{m}_0^2$ is an isomorphism, see Proposition 07E5. Now consider

$$\begin{array}{ccc} H_1(L_{K_0/k}) \otimes K & \longrightarrow & \mathfrak{m}_0/\mathfrak{m}_0^2 \otimes K \\ \downarrow & & \downarrow \\ H_1(L_{K/k}) & \longrightarrow & \mathfrak{m}/\mathfrak{m}^2 \end{array}$$

Since the left vertical arrow is injective by Lemma 07E2 and the lower horizontal by Proposition 07E5 we conclude that the right vertical one is too. Hence a regular system of parameters in A_0 maps to part of a regular system of parameters in A. We win by Algebra, Lemmas 07DY and 00NQ. □